

# Human Genetics III: Mendelian exceptions and Genomic Imprinting

Natarajan Ganesan, PhD  
Biomedical and Anatomic Sciences  
[nganesan@nyit.edu](mailto:nganesan@nyit.edu)

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Jonathan M Berman, PhD, Troy Camarata, PhD

# Learning Objectives

- Compare and contrast modes of non-Mendelian inheritance presented
- Differentiate between non-Mendelian and Mendelian modes of inheritance
- Interpret pedigrees of non-Mendelian disorders
- Relate anticipation to clinical presentation of dynamic mutations
- Identify clinical presentation of representative dynamic or genomic imprinting disorders
- Differentiate the genetic mechanisms that lead to dynamic and imprinting mutations
- Track parental origin of alleles involved in presented genomic imprinting disorders

# Additional References

- Medical Genetics  
<https://nyit.idm.oclc.org/login?url=https://www.clinicalkey.com/#!/content/book/3-s2.0-B9780323597371000060>
- Thompson and Thompson Genetics in Medicine  
<https://nyit.idm.oclc.org/login?url=https://www.clinicalkey.com/#!/browse/book/3-s2.0-C2009059798X>
- Emery's Elements of Medical Genetics  
<https://nyit.idm.oclc.org/login?url=https://www.clinicalkey.com/#!/browse/book/3-s2.0-C20180026808>
- Online Mendelian Inheritance of Man  
<https://www.ncbi.nlm.nih.gov/omim>

# Complex-USA level 1 blueprint- reference only

Information from this presentation relates to:

- REQUIRED ELEMENT 3.1
- FOUNDATIONAL BIOMEDICAL SCIENCES KNOWLEDGE BASE
  - the molecular, biochemical, tissue, and cellular bases of health and disease.
  - medical genetics.
- COMMUNITY HEALTH AND PATIENT PRESENTATIONS RELATED TO WELLNESS
  - genetic screening
- PATIENT PRESENTATIONS RELATED TO HUMAN DEVELOPMENT, REPRODUCTION, AND SEXUALITY
  - congenital/genetic anomalies (eg, Down syndrome, cystic fibrosis, craniosynostosis)
  - dysmorphic features and genetic syndrome stigmata

# From the USMLE Step 1 blueprint- reference only

Information from this presentation relates to:

- General Principles of Foundational Science Biochemistry and molecular biology
  - Gene expression: DNA structure, replication, exchange, and epigenetics (eg, imprinting, X-activation, DNA methylation)
- Human development and genetics
  - Principles of pedigree analysis inheritance patterns occurrence and recurrence risk determination
  - Population genetics: Hardy-Weinberg law, founder effects, mutation-selection equilibrium
  - Genetic testing and counseling, Genetic mechanisms (eg, penetrance, genetic heterogeneity)

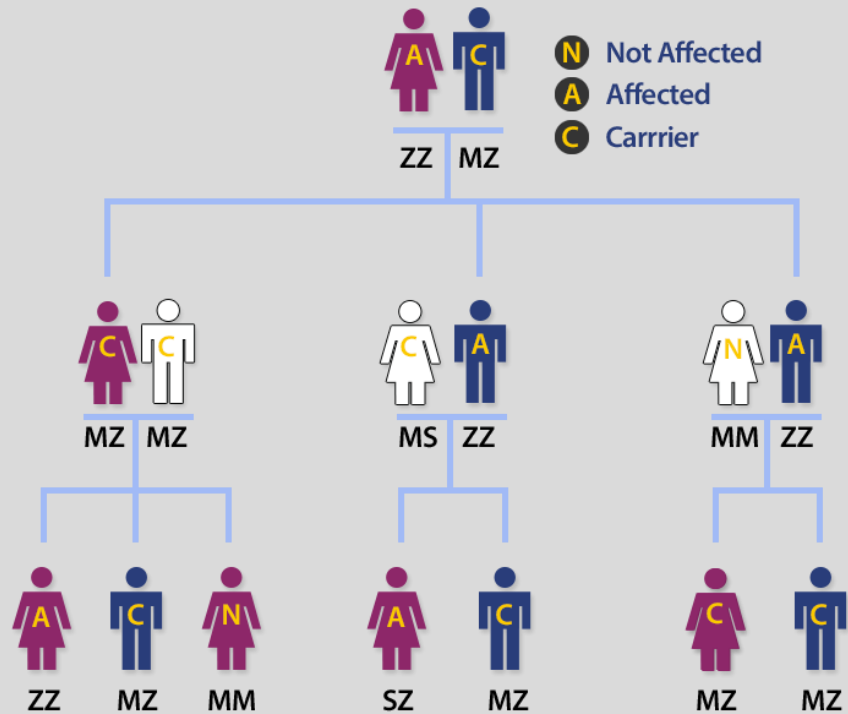
# From the USMLE Step 1 blueprint- reference only

Information from this presentation relates to:

- Abnormal Processes: Health and Health Maintenance, Screening, Diagnosis, Management, Risks, Prognosis
  - Genetic metabolic and developmental disorders multifactorial: VATER syndrome, association syndromes large genomic changes: Beckwith-Wiedemann syndrome; Down syndrome; Prader- Willi syndrome
  - enzymatic/metabolic: alpha-1 antitrypsin deficiency; porphyria; inborn errors of metabolism (eg, maple syrup urine disease, diseases involving urea cycle); storage diseases (eg, Fabry disease, Tay-Sachs disease, glycogen storage disease, mucopolysaccharidoses)
  - structural protein disorders: amyloidosis; Ehlers-Danlos syndrome; immotile cilia syndrome (Kartagener syndrome; primary ciliary dyskinesia); Marfan syndrome
  - intracellular/extracellular transport receptors: cystic fibrosis; hemochromatosis; Wilson disease
  - triplet repeat/RNA disorders: fragile X syndrome

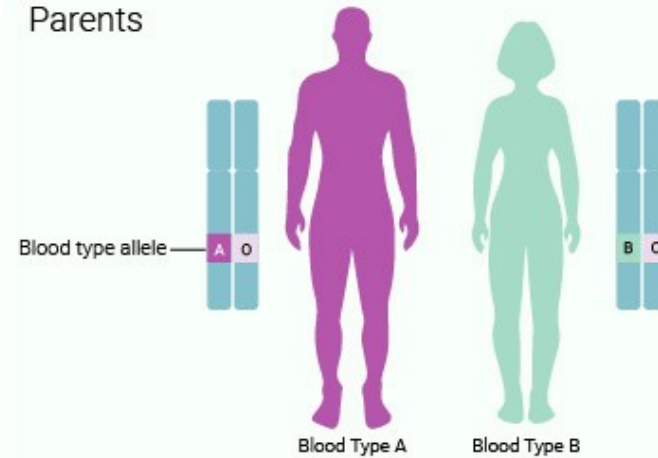
# Co-dominance

- Alpha-1 antitrypsin deficiency

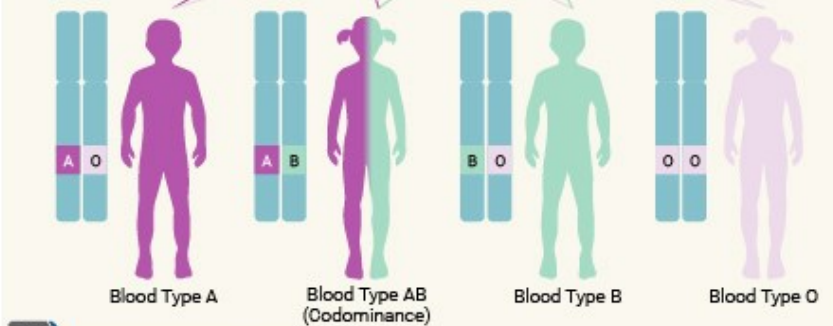


## Codominance - example Blood Type

Parents

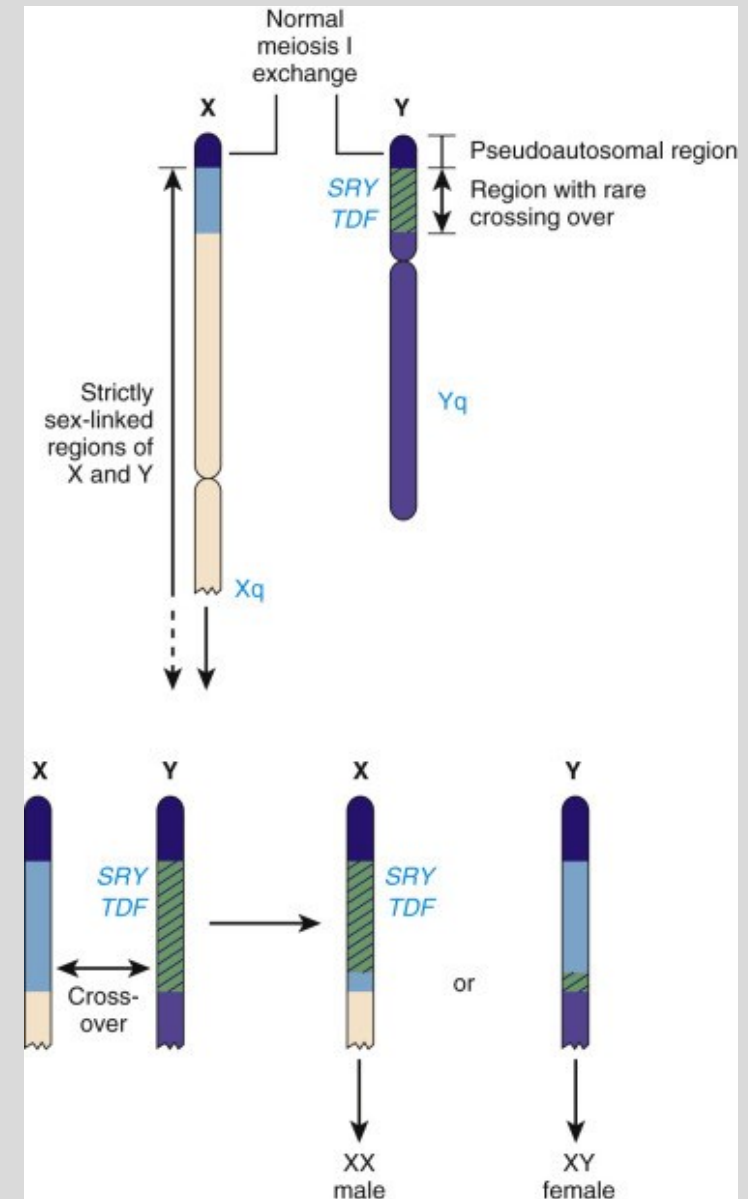
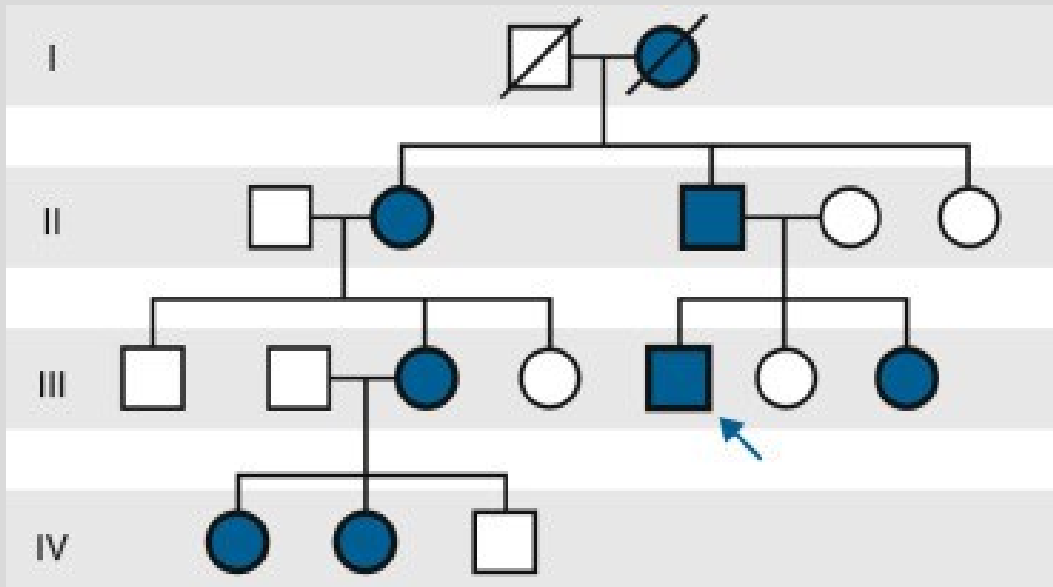


Children



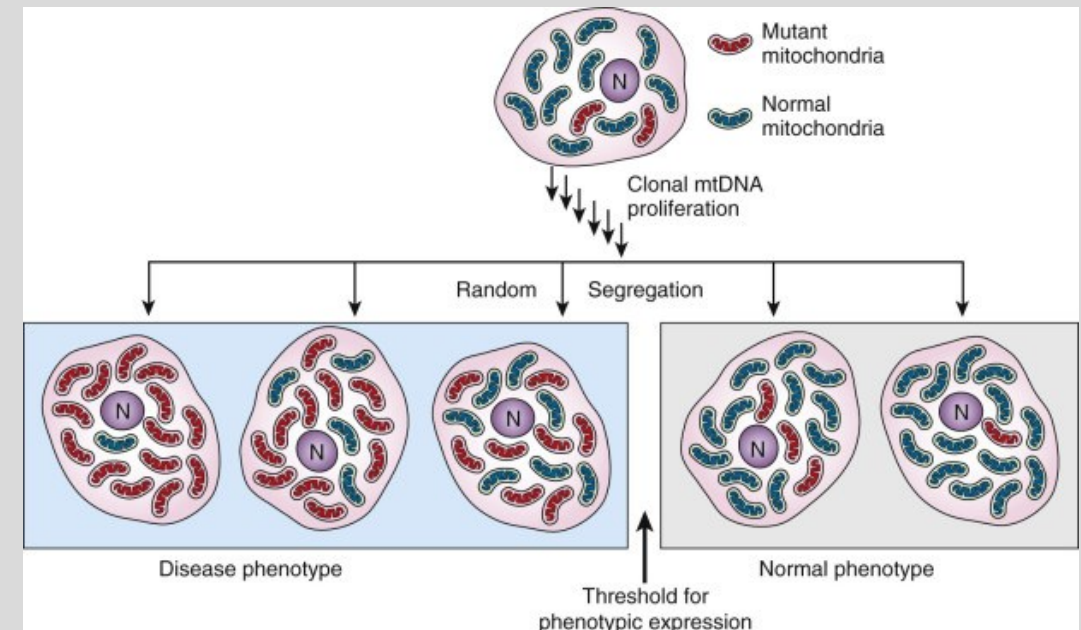
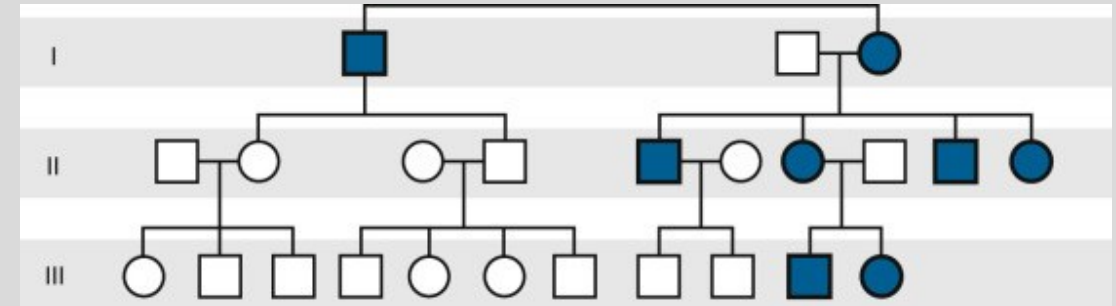
# Pseudoautosomal inheritance

- Limited homologous loci on X and Y
- Pseudoautosomal region
- Behave like autosomal genes
- Can get male to male inheritance



# Mitochondrial inheritance

- 37 genes in mtDNA
  - OxPhos
  - rRNA, tRNA
- Maternal inheritance
- Replicative segregation
  - mtDNA randomly distributed
- Homoplasmy, heteroplasmy
- Pleiotropic, reduced penetrance, variable expression



# Dynamic mutations

- Repeat expansion
  - Commonly trinucleotide repeat
- Wild-type alleles polymorphic
- Anticipation
  - Earlier disease onset with successive generations

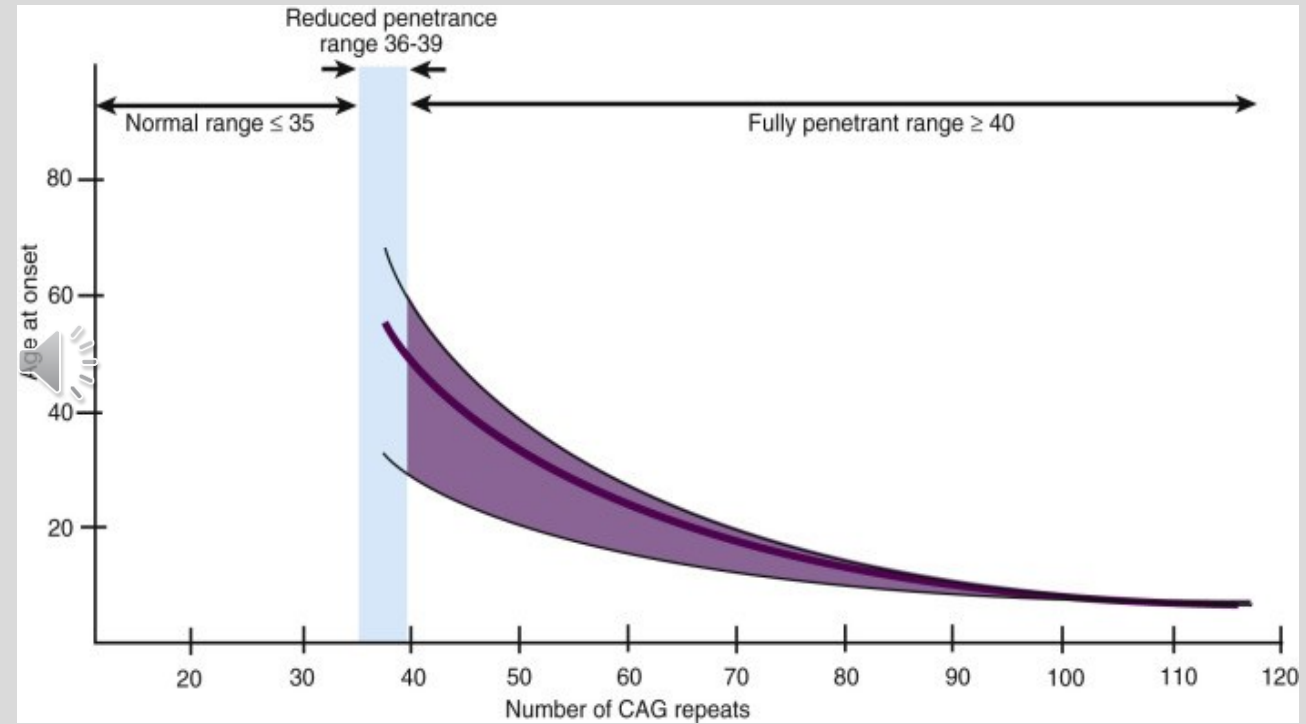


Fig. 7-20, Thompson & Thompson Genetics in Medicine, 8<sup>th</sup> Ed.

# Dynamic mutations- Examples

|                             | Huntington Disease             | Myotonic Dystrophy             |
|-----------------------------|--------------------------------|--------------------------------|
| <b>Inheritance</b>          | Autosomal dominant             | Autosomal dominant             |
| <b>Chromosome locus</b>     | 4p16.3                         | 19q13.3                        |
| <b>Trinucleotide repeat</b> | CAG in 5' translated region    | CTG in 3' untranslated region  |
| <b>Repeat sizes</b>         | Normal $\leq 26$               | Normal $< 37$                  |
|                             | Mutable 27–35                  |                                |
|                             | Reduced penetrance             | Full mutation                  |
|                             | 36–39                          | 50–2000+                       |
|                             | Fully penetrant $\geq 40$      |                                |
| <b>Protein product</b>      | Huntingtin                     | MD protein kinase (DMPK)       |
| <b>Early-onset form</b>     | Juvenile                       | Congenital                     |
|                             | Usually paternally transmitted | Usually maternally transmitted |

- Huntington disease: chorea, cognitive impairment, dementia, mood changes
- Myotonic dystrophy: progressive muscle weakness, myotonia, cataracts, cardiac conduction defects

# Fragile X syndrome

- 50-60% penetrance in females
  - 1:3000-4000 male births
- Most common form of inherited moderate intellectual disability
- CGG expansion
  - Normal up to 55 repeats
  - 56-200 premutation intermediate
  - >200 (up to several thousand) get disease
- Expansion leads to excessive methylation of *FMR1* promoter

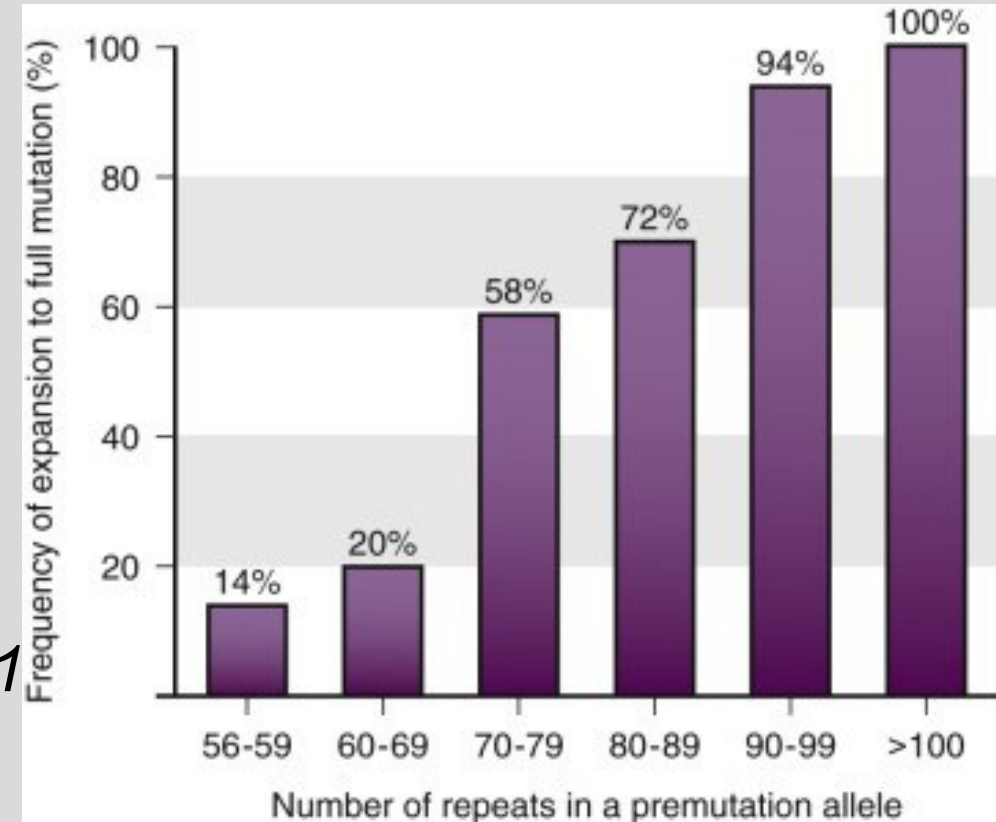


Fig. 7-23, Thompson & Thompson Genetics in Medicine, 8<sup>th</sup> Ed.

# Fragile X syndrome-characteristics

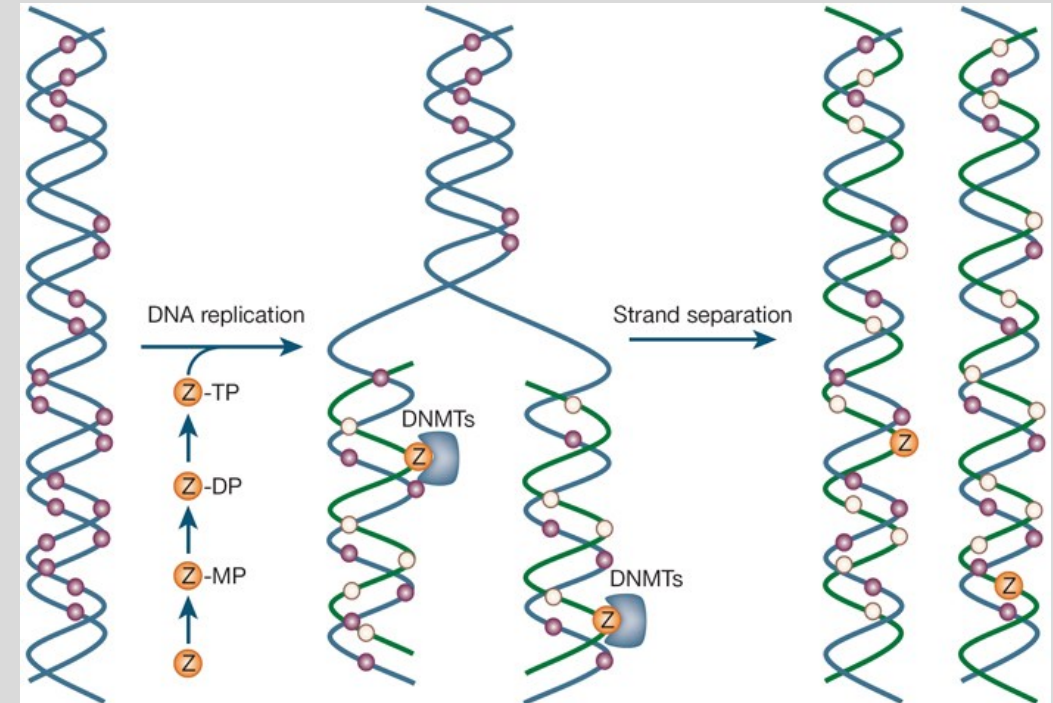
Physical presentations may include:

- Long face with long palpebral fissures as well as a broad philtrum
- Prominent forehead and protruding ears with soft cartilage
- High-arched palate and dental crowding
- Hyperextensible finger joints and thumbs
- Postpubertal macroorchidism 
- Hypotonia
- Biting, hand flapping, poor eye contact, language disorders from cluttered speech to complete lack of speech depending on phenotype severity
- Mild to profound intellectual disability
- Autism

<https://www.ncbi.nlm.nih.gov/books/NBK459243/>

# Epigenetics

- Influences gene regulation without altering DNA sequence
- Examples include: histone modification (chromatin structure), non-coding RNA (long non-coding RNA, microRNA), and DNA methylation (genomic imprinting)
- Play critical roles in embryonic development, congenital disease, and complex disease such as cancer



Dive in deeper if interested: <https://www.nature.com/scitable/topicpage/epigenetic-influences-and-disease-895/>  
<https://www.cdc.gov/genomics/disease/epigenetics.htm>

# Genomic imprinting

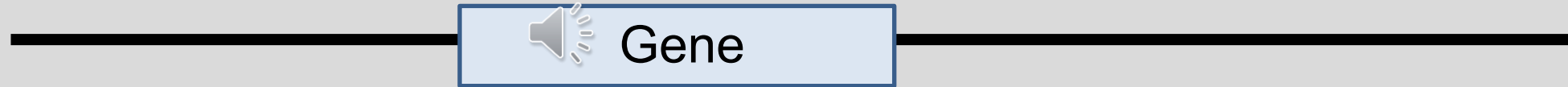
- Epigenetic phenomenon where genes are expressed as parent-of-origin specific manner
- Involves addition of chemical tags (e.g., DNA methylation) to the DNA during gamete formation – CpG islands
- Imprinting does not alter the DNA sequence itself but affects gene expression
- These modifications are established in the germline and maintained through mitotic cell divisions
- Examples:
  - IGF2 – only expressed from paternal allele
  - H19 – Expressed from maternal allele
- Crucial in embryonic development. Errors can lead to diseases like Angelman syndrome



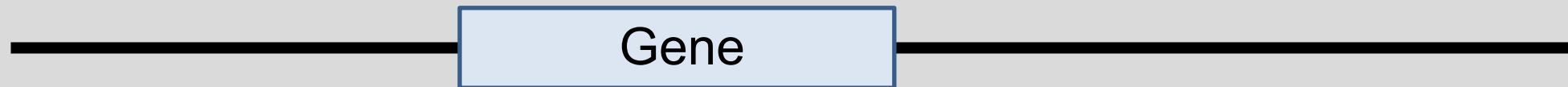
# How does this work?

- Genomic Imprinting:
  - **Tags** determine active parent gene.
- Tagging:
  - **Chemical markers** on genes.
- Parent-of-Origin:
  - **Only one parent's gene active.**
- Expression:
  - One gene expressed, other silent.
- **Crucial for development.**
- Example:
  - Paternal gene active, maternal silent.
- Importance:
  - Regulates development.

Paternal

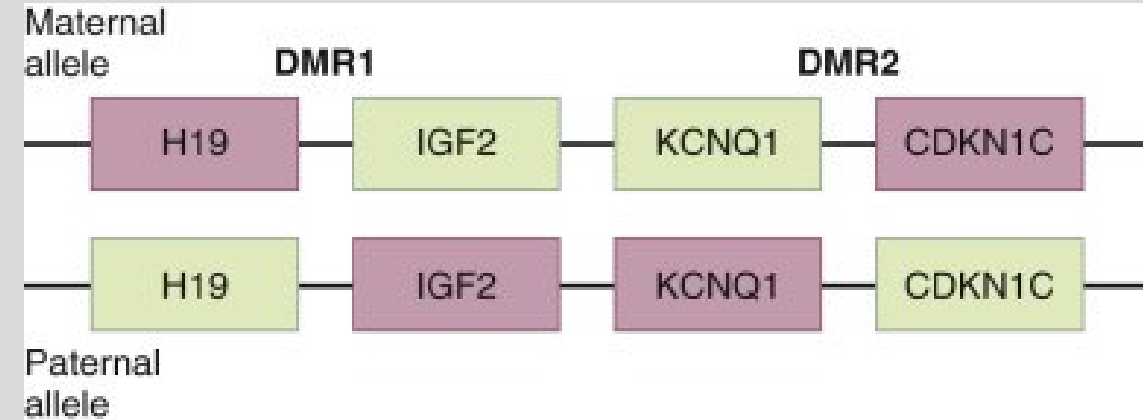


Maternal



# Beckwith-Wiedemann syndrome

- **Chromosome and Locations Affected:**
  - Chromosome 11p15.5
- **Key Genes**
  - **IGF2** – Overactivity promotes growth
  - **CDKN1C** – Loss of function leads to uncontrolled cell proliferation
  - **H19, KCNQ10T1** – Involved in growth regulation
- 20-30% uniparental paternal disomy
- 50-60% improper methylation maternally
- **Clinical features:**
  - **Overgrowth:** Macrosomia, hemihyperplasia
  - **Macroglossia:** Enlarged tongue
  - **Abdominal Wall Defects:** Omphalocele, umbilical hernia
  - **Hypoglycemia:** Low blood sugar in newborns
  - **Increased Cancer Risk:** Wilms tumor, hepatoblastoma
  - **Ear Anomalies:** Pits or creases in earlobe



Green alleles - methylated and off  
DMR: differential methylation region

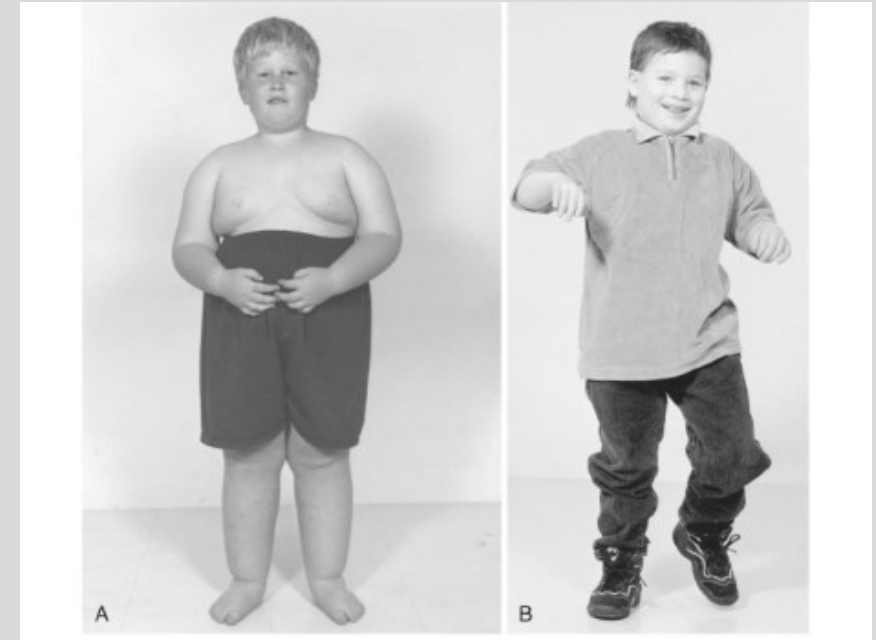
Fig 5-17 Medical Genetics, 5<sup>th</sup> Ed

- **Importance of Imprinting:**
  - **Genomic Imprinting:** Expression depends on parent of origin
  - **Epigenetic Regulation:** Disrupted DNA methylation and histone modifications

# Prader-Willi and Angelman Syndromes

- Small 4mb deletion 15q11.2-q13 (70%)
- Del in paternal 15q-PWS
- Del in maternal 15q-AS
- Most from de novo mutation

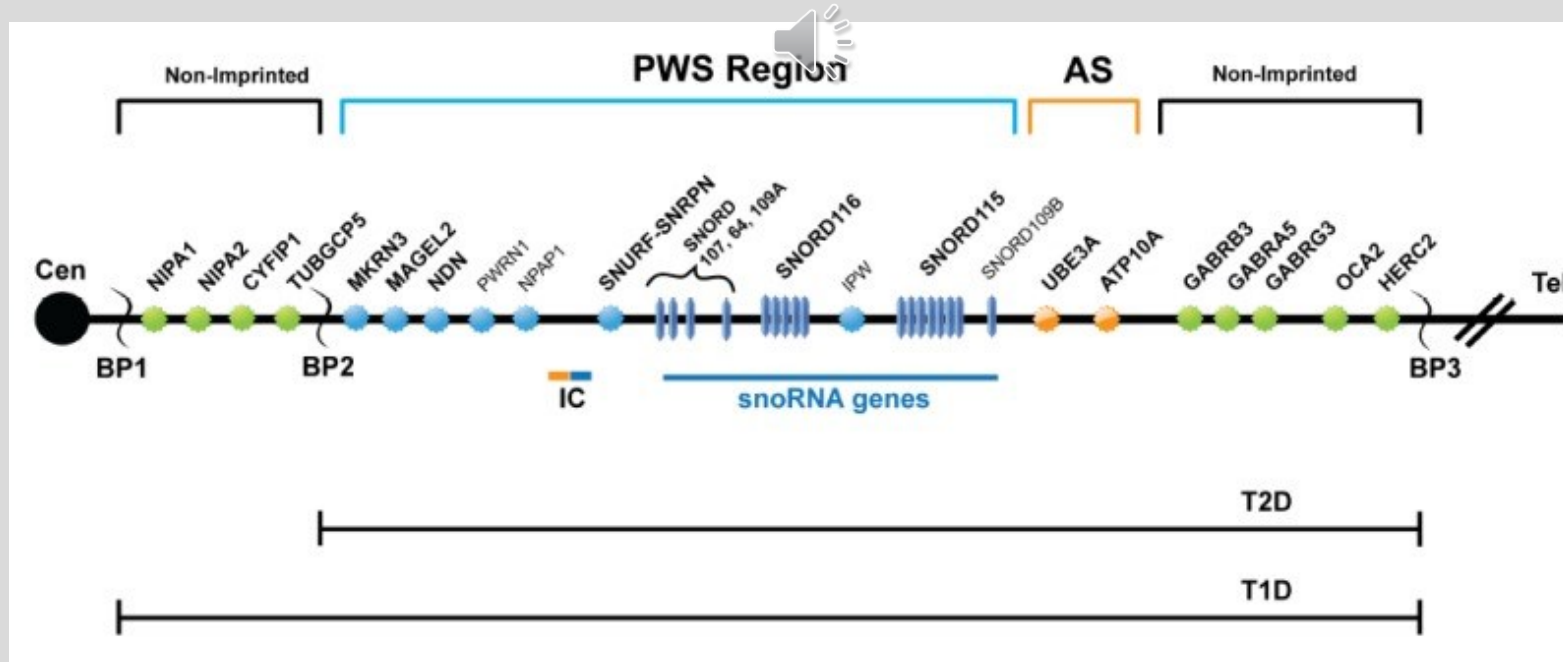
| Prader-Willi                          | Angelman                       |
|---------------------------------------|--------------------------------|
| 1:15,000                              | 1:15,000                       |
| short stature                         | short stature                  |
| neonatal hypotonia                    | seizures                       |
| small hands/feet                      | ataxic gait                    |
| hypogonadism                          | severe speech impairment       |
| obesity                               | microcephaly                   |
| mild/moderate intellectual disability | severe intellectual disability |
| developmental delay                   | developmental delay            |



A. Prader-Willi B. Angelman syndrome

# Mechanism of PWS and AS imprinting defect

- Small 4mb deletion 15q11.2-q13 (70%)
- Del in paternal 15q-PWS
- Del in maternal 15q-AS
  - Also from mutation in UBE3A gene



Cassidy et al, (2011). Prader-Willi Syndrome. *Genetics in Medicine* 14:10

# What would happen?

- Female with AS gives birth to female child? How about a male child?
- Male with AS fathers a female child? A male child?
- What would the offspring inherit and what would be the clinical manifestation?
- How is genetic counseling important here and what factors need to be considered?

A 7-year-old boy is brought to the clinic for obesity. He eats excessively and has increasing temper tantrums. His history is significant for infantile hypotonia with feeding difficulties and failure to thrive. He has had a delay in developmental milestones, and developmental testing reveals gross motor and intellectual delays; he is not verbal. Family history is noncontributory. Which of the following most likely accounts for this child's disorder?



- A. Absence of maternal gene expression
- B. Increased paternal gene expression
- C. Absence of paternal gene expression
- D. Increased mitochondrial homoplasmy
- E. Absence of X-linked gene expression

# Summary

- Recognize exceptions to Mendelian inheritance patterns
- Know and be able to identify representative diseases/syndromes
- Identify dynamic mutations and mechanisms behind anticipation
- Genomic imprinting and how  it impacts gene expression based upon parental origin
- Differentiate between AS and PWS and how these are inherited.
- What are the different mechanisms

<https://ghr.nlm.nih.gov/primer/inheritance/inheritancepatterns>

# Lecture Feedback Form:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>